

Module 16: Deployment & Hosting

Duration: 5-7 days | **Level:** Advanced

Hosting Options

Provider	Best For	Price
Laravel Forge	Managed VPS deployment	\$12+/mo
Laravel Vapor	Serverless (AWS Lambda)	Usage-based
DigitalOcean	VPS (Droplets)	\$6+/mo
AWS EC2 / Lightsail	Full control	Varies
Shared hosting	Simple apps (cPanel)	\$3+/mo
Railway / Render	Quick deploy from Git	Free tier

Server Requirements

- PHP 8.2+ with extensions (mbstring, openssl, pdo, tokenizer, xml, ctype, json, bcmath)
- Composer 2.x
- Nginx or Apache
- MySQL 8+ / PostgreSQL 15+ / SQLite (dev only)
- Redis (recommended for cache/queue)
- Node.js (build step only)

Deployment Checklist

1. Prepare Application

```
composer install --optimize-autoloader --no-dev
npm ci && npm run build
php artisan config:cache
php artisan route:cache
php artisan view:cache
php artisan event:cache
```

2. Environment

```
APP_ENV=production
APP_DEBUG=false
APP_URL=https://taskforge.example.com

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_DATABASE=taskforge
DB_USERNAME=taskforge
DB_PASSWORD=strong-password

CACHE_STORE=redis
QUEUE_CONNECTION=redis
SESSION_DRIVER=redis

MAIL_MAILER=smtp
MAIL_HOST=smtp.mailgun.org
```

3. Nginx Configuration

See taskforge/deploy/nginx.conf:

```
server {
    listen 80;
    server_name taskforge.example.com;
    root /var/www/taskforge/public;

    add_header X-Frame-Options "SAMEORIGIN";
    add_header X-Content-Type-Options "nosniff";

    index index.php;
    charset utf-8;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location ~ /\.php$ {
        fastcgi_pass unix:/var/run/php/php8.2-fpm.sock;
        fastcgi_param SCRIPT_FILENAME $realpath_root$fastcgi_script_name;
        include fastcgi_params;
    }

    location ~ /\.(!well-known).* {
        deny all;
    }
}
```

4. File Permissions

```
chown -R www-data:www-data /var/www/taskforge
chmod -R 755 /var/www/taskforge
chmod -R 775 storage bootstrap/cache
```

5. Run Migrations

```
php artisan migrate --force
php artisan storage:link
```

6. Process Managers

Queue worker (Supervisor):

```
[program:taskforge-worker]
process_name=%(program_name)s_%(process_num)02d
command=php /var/www/taskforge/artisan queue:work redis --sleep=3 --tries=3
--max-time=3600
autostart=true
autorestart=true
user=www-data
numprocs=2
```

Scheduler cron:

```
* * * * * www-data cd /var/www/taskforge && php artisan schedule:run >>
/dev/null 2>&1
```

SSL with Let's Encrypt

```
sudo apt install certbot python3-certbot-nginx
sudo certbot --nginx -d taskforge.example.com
```

Zero-Downtime Deployment

Use **Forge**, **Envoyer**, or **GitHub Actions** with:

1. Clone to new release directory
 2. `composer install --no-dev`
 3. `php artisan migrate --force`
 4. Symlink current -> new release
 5. Reload PHP-FPM
 6. Restart queue workers: `php artisan queue:restart`
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Docker / Laravel Sail

```
composer require laravel/sail --dev
php artisan sail:install
./vendor/bin/sail up
```

CI/CD with GitHub Actions

See [taskforge/deploy/github-actions.yml](#) for automated test + deploy pipeline.

Exercises

1. Deploy TaskForge to a free Render.com or Railway instance.
 2. Set up SSL and verify `https://` works.
 3. Configure Supervisor for queue workers.
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Next Module

-> [Module 17: Maintenance & Monitoring](#)